Laplace Transform (LT)

Definition of integral Transform / Kernel:

Let, $K(s,t)$ be a function of two variables s and t where s is a parameter (may be real or complex) independent of t. The function $f(s)$ defined by the integral (assumed to be convergent) $f(s) = \int_{-\infty}^{\infty} K(s,t)\, F(t)\, dt$ is called the integral transform of the function $F(t)$ and is denoted by $T\{F(t)\}$. The function $K(s,t)$ is sometimes called the kernel of the transformation.

Definition of Laplace transformation:

If the kernel $K(s,t)$ is defined as

$$K(s,t) = \begin{cases} 0 & \text{for } t<0 \\ e^{-st} & \text{for } t \geq 0 \end{cases}$$

then $f(s) = \int_0^{\infty} e^{-st} F(t)\, dt$ (1)

The function $f(s)$ defined by the integral (1) is called the Laplace Transform of the function $F(t)$ and is denoted by $\{F(t)\}$. Alternatively, let $F(t)$ be a function of t defined for all $t>0$.

Then the Laplace Transform of $F(t)$, denoted by $\mathcal{L}\{F(t)\}$ or $F(s)$ is defined by

$$\mathcal{L}\{F(t)\} = f(s) = \int_0^\infty e^{-st} F(t)\, dt$$

Table of Laplace Transform

$F(t)$	$\mathcal{L}\{F(t)\} = f(s)$
1	$\frac{1}{s}$, $s>0$
t	$\frac{1}{s^2}$, $s>0$
$\frac{t^n}{n!}$	$\frac{1}{s^{n+1}}$, $n=0,1,2,\ldots$
e^{at}	$\frac{1}{s-a}$, $s>a$
$\sin at$	$\frac{a}{s^2+a^2}$, $s>0$
$\cos at$	$\frac{s}{s^2+a^2}$, $s>0$
$\sinh at$	$\frac{a}{s^2-a^2}$, $s>\lvert a\rvert$
$\cosh at$	$\frac{s}{s^2-a^2}$, $s>\lvert a\rvert$

Find the Laplace Transform of the following elementary functions:

(i) $F(t) = 1$

We know that,

$$\mathcal{L}\{F(t)\} = f(s) = \int_0^\infty e^{-st} F(t)\,dt$$

$$\mathcal{L}\{F(t)\} = \int_0^\infty F(t)\, e^{-st}\,dt$$

$$= \int_0^\infty (1)\, e^{-st}\,dt$$

$$= \int_0^\infty e^{-st}\,dt = \left[-\frac{1}{s} e^{-st}\right]_0^\infty$$

$$= \left[-\frac{e^{-\infty}}{s} + \frac{1}{s}\cdot e^{0}\right] = \frac{1}{s}$$ (proved)

$e^{-\infty} = 0$

(ii) $F(t) = t$

$$\mathcal{L}\{F(t)\} = \int_0^\infty e^{-st} F(t)\,dt = \int_0^\infty e^{-st}\cdot t\,dt$$

$$\int uv = u\int v\,dx - \int\left\{\frac{d}{dx}(u)\int v\,dx\right\}dx$$

Here, $u = t$

$v = e^{-st}$

$$L\{F(t)\} = \int_0^\infty e^{-st} \cdot t\, dt$$

$$= t\int_0^\infty e^{-st}\, dt - \int_0^\infty \left\{\frac{d}{dt}(t)\int e^{-st}\, dt\right\} dt$$

$$= t\left[\frac{e^{-st}}{-s}\right]_0^\infty - \int_0^\infty 1.\frac{e^{-st}}{-s}\, dt$$

$$= 0 + \frac{1}{s}\int_0^\infty e^{-st}\, dt$$

$$= \frac{1}{s}\left[-\frac{1}{s}e^{-st}\right]_0^\infty$$

$$= \frac{1}{s}\left[-\frac{e^{-\infty}}{s} - \frac{e^0}{-s}\right]$$

$$= \frac{1}{s}\left(0 + \frac{1}{s}\right) = \frac{1}{s}\cdot\frac{1}{s} = \frac{1}{s^2}$$ (proved)

(III) $F(t) = t^n$

$$L\{F(t)\} = f(s) = \int_0^\infty e^{-st}\cdot F(t)\, dt$$

$$L\{t^n\} = \int_0^\infty e^{-st}\cdot t^n\, dt \quad \text{—(ii)}$$

Putting, $st = y$ so that $s\,dt = dy$ $\therefore dt = \frac{1}{s}dy$

Also, $t = \frac{y}{s}$

$\therefore t^n = \frac{y^n}{s^n}$ Limit, $t=0 \to t=\infty$, $y=0 \to y=\infty$

From (ii)

$$L\{t^n\} = \int_0^\infty e^{-y} \cdot \frac{y^n}{s^n} \cdot \frac{dy}{s}$$

$$= \frac{1}{s^{n+1}} \int_0^\infty e^{-y} \cdot y^n \cdot dy \quad \text{(Using gamma function)}$$

$$= \frac{1}{s^{n+1}} \boxed{\lceil n+1 \quad \text{(if } n > -1 \text{ and } s > 0\text{)}}$$

Again we know that, if n be a positive integer then $\lceil n+1 = n!$

Hence, $L\{t^n\} = \frac{n!}{s^{n+1}}$, when $n = 0, 1, 2, 3 \ldots$

(iv) $F(t) = e^{at}$

$$L\{e^{at}\} = \int_0^\infty e^{-st} \cdot e^{at} \cdot dt$$

$$= \int_0^\infty e^{-t(s-a)} \cdot dt$$

$$= -\frac{1}{s-a}\left[e^{-t(s-a)}\right]_0^\infty$$

$$= -\frac{1}{s-a} \cdot (0-1)$$

$$= \frac{1}{s-a} \quad \text{(proved)}$$

(v) $F(t) = \sin at$

$$L\{F(t)\} = \int_0^\infty e^{-st} \cdot F(t)\, dt$$

$$L\{\sin at\} = \int_0^\infty e^{-st} \cdot \sin at \cdot dt$$

$$= \left[\frac{e^{-st}}{s^2+a^2} (-s\sin at - a\cos at) \right]_0^\infty$$

$$= \left[-\frac{e^{-st}}{s^2+a^2}(s\sin at) - \frac{e^{-st}}{s^2+a^2}(a\cos at) \right]_0^\infty$$

$$= 0 + \frac{e^0}{s^2+a^2}(s\sin 0) + \frac{e^0}{s^2+a^2}(a\cos 0)$$

$$= 0 + 0 + \frac{1}{s^2+a^2} \times a = \frac{a}{s^2+a^2} \text{ (Proved)}$$

* $\int e^{ax} \sin bx \cdot dx = \frac{e^{ax}}{a^2+b^2}(a\sin bx - b\cos bx)$

* $\int e^{ax} \cos bx \cdot dx = \frac{e^{ax}}{a^2+b^2}(a\cos bx + b\sin bx)$

(vi) $F(t) = \cos at$

$$L\{\cos at\} = \int_0^\infty e^{-st} \cdot \cos at \cdot dt$$

$$= \left[\frac{e^{-st}}{s^2+a^2} (-s\cos at + a\sin at) \right]_0^\infty$$

$$= \left[-\frac{e^{-st}}{s^2+a^2}(s\cos at) + \frac{e^{-st}}{s^2+a^2}(a\sin at) \right]_0^\infty$$

$$= 0 + \frac{e^0}{s^2+a^2} \times s - \frac{e^0}{s^2+a^2} \times 0$$

$$= \frac{s}{s^2+a^2} \text{ (Proved)}$$

(vii) $F(t) = \sinh at$

$$L\{\sinh at\} = \int_0^\infty e^{-st} \sinh at \, dt$$

$$= \int_0^\infty e^{-st} \cdot \frac{1}{2}(e^{at} - e^{-at}) \, dt \quad \left| \sinh at = \frac{1}{2}(e^t - e^{-t}) \right.$$

$$= \frac{1}{2}\int_0^\infty e^{-st} \cdot e^{at} \, dt - \frac{1}{2}\int_0^\infty e^{-st} e^{-at} \, dt$$

$$= \frac{1}{2}\int_0^\infty e^{-t(s-a)} \, dt - \frac{1}{2}\int_0^\infty e^{-t(s+a)} \, dt$$

$$= \frac{1}{2}\left[\frac{e^{-t(s-a)}}{-(s-a)}\right]_0^\infty + \frac{1}{2}\left[\frac{e^{-t(s+a)}}{-(s+a)}\right]_0^\infty$$

$$= \frac{1}{2}\left[\frac{e^{-t(s-a)}}{-(s-a)}\right]_0^\infty + \frac{1}{2}\left[\frac{e^{-t(s+a)}}{s+a}\right]_0^\infty$$

$$= -\frac{1}{2(s-a)}\left[e^{-t(s-a)}\right]_0^\infty + \frac{1}{2(s+a)}\left[e^{-t(s+a)}\right]_0^\infty$$

$$= -\frac{1}{2(s-a)}(0-1) + \frac{1}{2(s+a)}(0-1)$$

$$= \frac{1}{2(s-a)} - \frac{1}{2(s+a)}$$

$$= \frac{s+a-s+a}{2(s-a)(s+a)} = \frac{2a}{2(s^2-a^2)} = \frac{a}{s^2-a^2}$$

(Proved)

Ans

(viii) $F(t) = \cosh at$

$\cosh at = \frac{1}{2}(e^{t} + e^{-t})$

$$L\{\cosh at\} = \int_0^\infty e^{-st} \cdot \cosh at \, dt$$

$$= \int_0^\infty e^{-st} \cdot \frac{1}{2}(e^{at} + e^{-at}) \, dt$$

$$= \frac{1}{2}\int_0^\infty e^{-st} \cdot e^{at} \, dt + \frac{1}{2}\int_0^\infty e^{-st} \cdot e^{-at} \, dt$$

$$= \frac{1}{2}\int_0^\infty e^{-t(s-a)} \, dt + \frac{1}{2}\int_0^\infty e^{-t(s+a)} \, dt$$

$$= \frac{1}{2}\left[\frac{e^{-t(s-a)}}{-(s-a)}\right]_0^\infty + \frac{1}{2}\left[\frac{e^{-t(s+a)}}{-(s+a)}\right]_0^\infty$$

$$= -\frac{1}{2(s-a)}\left[e^{-t(s-a)}\right]_0^\infty + \frac{1}{2(s+a)}\left[e^{-t(s+a)}\right]_0^\infty$$

$$= -\frac{1}{2(s-a)}(0-1) - \frac{1}{2(s+a)}(0-1)$$

$$= \frac{1}{2(s-a)} + \frac{1}{2(s+a)}$$

$$= \frac{s+a+s-a}{2(s-a)(s+a)}$$

$$= \frac{2s}{2(s^2-a^2)} = \frac{s}{s^2-a^2}$$ (Proved)

Linearity Property

Formula:

$$\mathcal{L}\{c_1 F_1(t) + c_2 F_2(t)\} = c_1 \mathcal{L}\{F_1(t)\} + c_2 \mathcal{L}\{F_2(t)\}$$
$$= c_1 f_1(s) + c_2 f_2(s)$$

Example:

(1) Find $\mathcal{L}\{5e^{2t} - 3\}$

$= \mathcal{L}\{5e^{2t}\} - \mathcal{L}\{3\}$

$= 5\mathcal{L}\{e^{2t}\} + 3\mathcal{L}\{1\}$

$= \frac{5}{s-2} - \frac{3}{s}$ Ans

$\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$ (i)

$\mathcal{L}\{1\} = \frac{1}{s}$

(2) $\mathcal{L}\{6\sin 2t - 5\cos 2t\}$

$= \mathcal{L}\{6\sin 2t\} - \mathcal{L}\{5\cos 2t\}$

$= 6\mathcal{L}\{\sin 2t\} - 5\mathcal{L}\{\cos 2t\}$

$= 6 \cdot \frac{2}{s^2+4} - 5 \cdot \frac{s}{s^2+4} = \frac{12}{s^2+4} - \frac{5s}{s^2+4}$ Ans

$\mathcal{L}\{\sin at\} = \frac{a}{s^2+a^2}$

$\mathcal{L}\{\cos at\} = \frac{s}{s^2+a^2}$

(iii) $\mathcal{L}\{4e^{5t} + 6t^3 - 3\cos 4t + 4\sin 5t\}$

$= \mathcal{L}\{4e^{5t}\} + \mathcal{L}\{6t^3\} - \mathcal{L}\{3\cos 4t\} + \mathcal{L}\{4\sin 5t\}$

$= 4\mathcal{L}\{e^{5t}\} + 6\mathcal{L}\{t^3\} - 3\mathcal{L}\{\cos 4t\} + 4\mathcal{L}\{\sin 5t\}$

$= 4 \cdot \frac{1}{s-5} + 6\mathcal{L}\{t^3\} - 3 \cdot \frac{s}{s^2+16} + 4 \cdot \frac{5}{s^2+25}$

$= \frac{4}{s-5} + 6 \cdot \mathcal{L}\{t^3\} - \frac{3s}{s^2+16} + \frac{20}{s^2+25}$

$= \frac{4}{s-5} + 6 \cdot \frac{3!}{s^{3+1}} - \frac{3s}{s^2+16} + \frac{20}{s^2+25}$

$= \frac{4}{s-5} + \frac{36}{s^4} - \frac{3s}{s^2+16} + \frac{20}{s^2+25}$ Answer

(4) Find the Laplace Transform of the following functions.

(i) $F(t) = t\sin at$ (ii) $F(t) = t\cos at$

⇒ By definition of the Laplace transform of a function $F(t)$, we have

$$L\{F(t)\} = f(s) = \int_0^\infty e^{-st} F(t)\, dt \quad \text{---(1)}$$

(i) When $F(t) = t\sin at$, equation (1) becomes

$$L\{t\sin at\} = \int_0^\infty e^{-st} \cdot t\sin at\, dt$$

$$= \int_0^\infty t \cdot e^{-st} \sin at\, dt$$

Integrating by parts, taking t as first function and $e^{-st}\sin at$ as the second function and applying the result.

$$\int e^{ax} \sin bx \, dx = \frac{e^{ax}}{a^2+b^2}\left[a\sin bx - b\cos bx\right]$$

$$\therefore L\{t\sin at\} = \int_0^\infty t \cdot e^{-st} \sin at\, dt$$

$$\left[\int uv = u\int v\,dx - \int\left\{\frac{d}{dx}(u)\int v\,dx\right\} dx\right.$$

$$= t\int_0^\infty e^{-st}\sin at\, dt - \int_0^\infty \left\{\frac{d}{dt}(t)\int e^{-st}\sin at\right\} dt$$

$$= \left[t \cdot \frac{e^{-st}}{s^2+a^2}\left(-s\sin at - a\cos at\right)\right]_0^\infty$$

$$- \int_0^\infty e^{-st} \cdot \sin at\, dt$$

$$= 0 - \left\{ \int_0^\infty \frac{e^{-st}}{s^2+a^2} (a\sin at - a\cos at) \right\}_0^\infty dt$$

$$= \int_0^\infty \frac{e^{-st}}{s^2+a^2} (s\sin at + a\cos at)\, dt$$

$$= \int_0^\infty \frac{e^{-st}}{s^2+a^2} \cdot s\sin at\, dt + \int_0^\infty \frac{e^{-st}}{s^2+a^2} \cdot a\cos at\, dt$$

$$= \frac{s}{s^2+a^2} \int_0^\infty e^{-st} \sin at\, dt + \frac{a}{s^2+a^2} \int_0^\infty e^{-st} \cos at\, dt$$

$$= \frac{s}{s^2+a^2} \left[\frac{e^{-st}}{s^2+a^2} (-s\sin at - a\cos at) \right]_0^\infty$$

$$+ \frac{a}{s^2+a^2} \left[\frac{e^{-st}}{s^2+a^2} (-s\cos at + a\sin at) \right]_0^\infty$$

$$= \frac{s}{s^2+a^2} \left(0 - \frac{e^0}{s^2+a^2} \cdot -a\cos 0 \right) + \frac{a}{s^2+a^2} \left(0 - \frac{e^0}{s^2+a^2} \cdot -s\cos 0 \right)$$

$$= \frac{s}{s^2+a^2} \times \frac{a}{s^2+a^2} + \frac{a}{s^2+a^2} \times \frac{s}{s^2+a^2}$$

$$= \frac{2as}{(s^2+a^2)^2} + \frac{as}{(s^2+a^2)^2}$$

$$= \frac{2as}{(s^2+a^2)^2} \quad \text{for } s > 0$$

(Answer)

(ii) When $F(t) = t\cos at$, equation (1) becomes

$$L\{t\cos at\} = \int_0^\infty e^{-st} \cdot t\cos at \, dt$$

$$= \int_0^\infty t \cdot e^{-st} \cos at \, dt$$

$$\left[\int uv = u\int v\,dx - \int\left\{\frac{d}{dx}(u)\int v\,dx\right\}dx\right]$$

Integrating by parts, taking t as first function and $e^{-st}\cdot\cos at$ as the second function and applying the result.

$$\left[\int e^{ax}\cdot\cos bx\,dx = \frac{e^{ax}}{a^2+b^2}\left[a\cos bx + b\sin bx\right]\right]$$

we get,

$$L\{t\cos at\} = \left[t\cdot\frac{e^{-st}}{s^2+a^2}(-s\cos at + a\sin at)\right]_0^\infty$$

$$- \int_0^\infty \left\{\frac{d}{dt}(t)\int e^{-st}\cdot\cos at\,dt\right\}$$

$$= 0 - \int_0^\infty e^{-st}\cos at\,dt$$

$$= -\int_0^\infty \frac{e^{-st}}{s^2+a^2}(-s\cos at + a\sin at)\,dt$$

$$= \int_0^\infty \frac{e^{-st}}{s^2+a^2}(s\cos at - a\sin at)\,dt$$

$$= \int_0^\infty \frac{e^{-st}}{s^2+a^2} \cdot s\cos at\,dt \int_0^\infty \frac{e^{-st}}{s^2+a^2} \cdot a\sin at\,dt$$

$$= \frac{s}{s^2+a^2}\int_0^\infty e^{-st}\cdot \cos at\,dt - \frac{a}{s^2+a^2}\int_0^\infty e^{-st}\sin at\,dt$$

$$= \frac{s}{s^2+a^2}\left[\frac{e^{-st}}{s^2+a^2}(-s\cos at + a\sin at)\right]_0^\infty$$

$$- \frac{a}{s^2+a^2}\left[\frac{e^{-st}}{s^2+a^2}(-s\sin at - a\cos at)\right]_0^\infty$$

$$= \left(\frac{s}{s^2+a^2}\times - \frac{e^0}{s^2+a^2}\cdot -s\right) - \left(\frac{a}{s^2+a^2}\times \frac{e^0}{s^2+a^2}\cdot -a\right)$$

$$= \frac{s^2}{(s^2+a^2)^2} + \frac{a^2}{(s^2+a^2)^2}$$

$$= \frac{s^2+a^2}{(s^2+a^2)^2} \quad \text{for } s>0$$

$$= \frac{1}{s^2+a^2}$$

(Answer)

Laplace Transform of derivatives

* If $L\{F(t)\} = f(s)$ then

$$\left[L\{F''(t)\} = s^2 f(s) - sF(0) - F'(0)\right] \text{ [Formula]}$$

Find the Laplace transform of $\cos at$ using derivatives.

Solution: Let, $F(t) = \cos at$

$$F'(t) = -a\sin at$$

$$F''(t) = -a^2 \cos at$$

$$F(0) = \cos 0 = 1$$

$$F'(0) = -a\sin 0 = 0$$

We know,

$$L\{F''(t)\} = s^2 f(s) - sF(0) - F'(0)$$

or, $L\{-a^2 \cos at\} = s^2 \cdot L\{\cos at\} - s.1 - 0$

or, $-a^2 L\{\cos at\} = s^2 \cdot L\{\cos at\} - s$

or, $(s^2 + a^2) L\{\cos at\} = s$

$\therefore L\{\cos at\} = \dfrac{s}{s^2 + a^2}$

(Answer)

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Find the Laplace transform of sinat using derivatives.

Solution:

Let, $F(t) = \sin at$

$F'(t) = a\cos at$

$F''(t) = -a^2 \sin at$

$F(0) = 0$

$F'(0) = a$

We know,

$$L\{F''(t)\} = s^2 f(s) - sF(0) - F'(0)$$

or, $L\{-a^2 \sin at\} = s^2 L\{\sin at\} - 0 - a$

or, $-a^2 L\{\sin at\} = s^2 L\{\sin at\} - a$

or, $(s^2 + a^2) L\{\sin at\} = a$

$$\therefore L\{\sin at\} = \frac{a}{s^2 + a^2}$$ (Answer)

First derivative :- $L\{F'(t)\} = sf(s) - F(0)$

2nd $L\{F'''(t)\} = s^3 sf(s) - s^2 f(0) - sF''(0) - F''(0)$

Second Translation (or shifting) Property:

Theorem: If $L\{F(t)\} = f(s)$ and $G_2(t) = \begin{cases} F(t-a), & t > a \\ 0, & t < a \end{cases}$

then $L\{G_2(t)\} = e^{-as} f(s)$.

Proof: By definition of the Laplace transform we have,

$$L\{F(t)\} = \int_0^\infty e^{-st} F(t)\, dt = f(s)$$

$$\therefore L\{G_2(t)\} = \int_0^\infty e^{-st} G_2(t)\, dt$$

$$= \int_0^a e^{-st} G_2(t)\, dt + \int_a^\infty e^{-st} G_2(t)\, dt$$

$$= \int_0^a e^{-st}(0)\, dt + \int_a^\infty e^{-st} F(t-a)\, dt$$

$$= 0 + \int_a^\infty e^{-st} F(t-a)\, dt \quad \text{—(i)}$$

Let, $t - a = v \quad \therefore dt = dv$

Limits, $t = a \Rightarrow v = 0$, $t = \infty \Rightarrow v = \infty$

Thus from (i), we have

$$L\{G_2(t)\} = \int_0^\infty e^{-s(v+a)} \cdot F(v)\, dv$$

$$= e^{-sa} \int_0^\infty e^{-sv} \cdot F(v)\, dv$$

$$= e^{-sa} f(s)$$

The change of scale property:

Theorem: If $L\{F(t)\} = f(s)$ then $L\{F(at)\} = \frac{1}{a} f\left(\frac{s}{a}\right)$

Proof – By definition of the Laplace Transform we have,

$$L\{F(t)\} = \int_0^\infty e^{-st} F(t)\, dt = f(s)$$

$$\therefore L\{F(at)\} = \int_0^\infty e^{-st} F(at)\, dt \quad \text{---(i)}$$

Let, $at = u$ or, $t = \frac{u}{a}$ $\therefore dt = \frac{1}{a} du$

Limits, $t = 0 \Rightarrow u = 0$ and $t = \infty \Rightarrow u = \infty$

Thus from (i) we have

$$L\{F(at)\} = \int_0^\infty e^{-\frac{su}{a}} \cdot F(u) \cdot \frac{1}{a} du$$

$$= \frac{1}{a} \int_0^\infty e^{\left(-\frac{s}{a}\right)u} \cdot F(u)\, du$$

$$= \frac{1}{a} f\left(\frac{s}{a}\right)$$

(proved)

Derivatives of Laplace Transforms:

(Multiplication by powers of t)

* If $L\{F(t)\} = f(s)$ then

Theorem:

$$L\{t^n F(t)\} = (-1)^n \frac{d^n}{ds^n} f(s) = (-1)^n f^n(s)$$

where, $n = 1, 2, 3 \ldots$

Proof: By definition of the Laplace transform we have

$$f(s) = \int_0^\infty e^{-st} F(t)\, dt$$

Now applying Leibnitz's rule for differentiating under the integral sign we get,

$$\frac{df}{ds} = f'(s) = \frac{d}{ds} \int_0^\infty e^{-st} F(t)\, dt$$

$$= \int_0^\infty \frac{\partial}{\partial s} e^{-st} F(t)\, dt$$

$$= \int_0^\infty -t e^{-st} F(t)\, dt$$

$$= \int_0^\infty -e^{-st} \{t F(t)\}\, dt$$

$$= -L\{t F(t)\}.$$

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Thus, $L\{tF(t)\} = -\frac{dF}{ds} = -f'(s)$ —— (i)

Hence the theorem is proved for $n=1$

To prove the theorem in general, let us use the mathematical induction, suppose that the theorem is true for $n=k$. Then we have

$$\int_0^\infty e^{-st}\{t^k F(t)\}\,dt = (-1)^k f^k(s) \quad \text{—— (ii)}$$

Differentiating both sides with respect to s, we get

$$\frac{d}{ds}\int_0^\infty e^{-st}\{t^k F(t)\}\,dt = (-1)^k f^{k+1}(s)$$

Now, applying Leibnitz's rule we get,

$$-\int_0^\infty e^{-st}\{t^{k+1} F(t)\}\,dt = (-1)^k f^{k+1}(s)$$

or, $$-\int_0^\infty e^{-st}\{t^{k+1} F(t)\}\,dt = (-1)^{k+1} f^{k+1}(s) \quad \text{—— (iii)}$$

Thus it follows that if the theorem holds for $n=k$ then it also holds for $n=k+1$.

But in (1) it is proved that the theorem holds for $n=1$. Hence it holds for $n=1+1=2$ and $n=2+1=3$ etc. Thus, the theorem holds for all positive integral values of n.

$$L\{t^n F(t)\} = (-1)^n \frac{d^n}{ds^n} f(s) = (-1)^n f^n(s)$$ (proved)

Example:

① Find $L\{t\sin at\}$

We know that,

$$L\{F(t)\} = L\{\sin at\} = \frac{a}{s^2+a^2} = f(s)$$

where, $F(t) = \sin at$

Again, $L\{t F(t)\} = L\{t\sin at\}$

$$= -\frac{d}{ds} f(s)$$

$$= -\frac{d}{ds}\left(\frac{a}{s^2+a^2}\right)$$

$$= -\frac{d}{ds}\left(\frac{a}{s^2}+\frac{1}{a}\right)$$

$$= -\frac{d}{ds}\left(\frac{a}{s^2}\right) - \frac{d}{ds}\left(\frac{1}{a}\right)$$

$$= \frac{2a}{s^3}$$ (Answer)

(11) Find $L\{t^2 \cos at\}$:

We know that,

$$L\{F(t)\} = L\{\cos at\} = \frac{s}{s^2+a^2} \quad \text{---(1)}$$

Where, $F(t) = \cos at$

Again, $L\{t^2 F(t)\} = L\{t^2 \cos at\}$

$$= \frac{d^2}{ds^2}\left(\frac{s}{s^2+a^2}\right)$$

First differentiation:

$$\frac{d}{ds}\left(\frac{s}{s^2+a^2}\right) \quad \left[\frac{d}{dx}(uv) = u\frac{d}{dx}(v) + v\frac{d}{dx}\right.$$

$$= \frac{d}{ds}\left(\frac{s \cdot 1}{s} + \frac{s}{a^2}\right) = \frac{d}{ds}$$

$$= s\frac{d}{ds}(s^2+a^2)^{-1} + (s^2+a^2)^{-1}\frac{d}{ds}(s)$$

$$= -s\cdot(s^2+a^2)^{-2}\cdot 2s + (s^2+a^2)^{-1}$$

$$= \frac{-2s^2}{(s^2+a^2)^2} + \frac{1}{s^2+a^2}$$

$$= \frac{-2s^2+s^2+a^2}{(s^2+a^2)^2}$$

$$= \frac{a^2-s^2}{(s^2+a^2)^2}$$

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Second differentiation:

$$\frac{d}{ds}\left[\frac{a^2-s^2}{(s^2+a^2)^2}\right]$$

$$= \frac{d}{ds}\left[(a^2-s^2)(s^2+a^2)^{-2}\right]$$

$$= (a^2-s^2)\frac{d}{ds}(s^2+a^2)^{-2} + (s^2+a^2)^{-2}\frac{d}{ds}(a^2-s^2)$$

$$= (a^2-s^2).-2(s^2+a^2)^{-3}.2s + (s^2+a^2)^{-2}.-2s$$

$$= \frac{-4s(a^2-s^2)}{(s^2+a^2)^3} - \frac{2s}{(s^2+a^2)^2}$$

$$= \frac{-4s(a^2-s^2) - 2s(s^2+a^2)}{(s^2+a^2)^3}$$

$$= \frac{-4sa^2+4s^3-2s^3-2a^2s}{(s^2+a^2)^3}$$

$$= \frac{2s^3-6a^2s}{(s^2+a^2)^3}$$

$$= \frac{2s(s^2-3a^2)}{(s^2+a^2)^3}$$

(Answer)

(iii) Find $L\{t(\sin 3t + e^{2t})\}$

$L\{F(t)\} = L\{\sin 3t\} = \frac{3}{s^2+9}$ Where, $F(t) = \sin 3t$

$L\{\underset{F(t)}{e^{2t}}\} = L\{e^{2t}\} = \frac{1}{s-2}$, where $F(t) = e^{2t}$

$L\{t(\sin 3t + e^{2t})\} = -\frac{d}{ds}\left[\frac{3}{s^2+9} + \frac{1}{s-2}\right]$

Differentiate:

$$\frac{d}{ds}\left(\frac{3}{s^2+9}\right) = \frac{d}{ds}\{3(s^2+9)^{-1}\}$$

$$= -3(s^2+9)^{-2} \cdot 2s = -\frac{6s}{(s^2+9)^2}$$

$$\frac{d}{ds}\left(\frac{1}{s-2}\right) = \frac{d}{ds}(s-2)^{-1} = -1(s-2)^{-2} \cdot 1$$

$$= -\frac{1}{(s-2)^2}$$

$$\therefore -\frac{d}{ds}\left[\frac{3}{s^2+9} + \frac{1}{s-2}\right] = -\left[-\frac{6s}{(s^2+9)^2} - \frac{1}{(s-2)^2}\right]$$

$$= \frac{6s}{(s^2+9)^2} + \frac{1}{(s-2)^2}$$

(Answer)

Worked Out Examples

Example-1:- Find the Laplace Transform of $e^{4t}+4t^3-2\sin 3t+3\cos 5t$

$\Rightarrow$ By applying Linear property, we have

$$L\{e^{4t}+4t^3-2\sin 3t+3\cos 5t\}$$

$$=L\{e^{4t}\}+4L\{t^3\}-2L\{\sin 3t\}+3L\{\cos 5t\}$$

$$=\frac{1}{s-4}+4.\frac{L^3}{s^4}-2.\frac{3}{s^2+9}+3.\frac{s}{s^2+25}$$

$$=\frac{1}{s-4}+\frac{4\times 6}{s^4}-\frac{6}{s^3+9}+\frac{3s}{s^2+25}$$

$$=\frac{1}{s-4}+\frac{24}{s^4}-\frac{6}{s^3+9}+\frac{3s}{s^2+25}$$ (Answer)

Example-2: Find the Laplace transform of $3t^4-2t^3+4e^{-3t}-2\sin 5t+3\cos 2t$

$\Rightarrow$ By applying Linear property, we have

$$L\{3t^4-2t^3+4e^{-3t}-2\sin 5t+3\cos 2t\}$$

$$=3L\{t^4\}-2L\{t^3\}+4L\{e^{-3t}\}-2L\{\sin 5t\}+3L\{\cos 2t\}$$

$$=3.\frac{L4}{s^5}-2.\frac{L3}{s^4}+4.\frac{1}{s+3}-2.\frac{5}{s^2+25}+3.\frac{s}{s^2+4}$$

$$=\frac{72}{s^5}-\frac{12}{s^4}+\frac{4}{s+3}-\frac{10}{s^2+25}+\frac{3s}{s^2+4}$$

(Answer)

Example-3 Find the Laplace Transform of t^3e^{5t}

$\Rightarrow \mathcal{L}\{e^{5t}\} = \frac{1}{s-5}$

Applying multiplication property

$$\mathcal{L}\{t^3e^{5t}\} = (-1)^3 \cdot \frac{d^3}{ds^3}\left(\frac{1}{s-5}\right)$$

$$= -\frac{d^3}{ds^3}\left(\frac{1}{s-5}\right)$$

First differentiation

$$\frac{d}{ds}\left(\frac{1}{s-5}\right) = -\frac{d}{ds}(s-5)^{-1} = (s-5)^{-2}$$

2nd differentiation

$$\frac{d^2}{ds^2} = \frac{d}{ds}(s-5)^{-2} = -2(s-5)^{-3}$$

3rd differentiation

$$\frac{d^3}{ds^3} = \frac{d}{ds}\left(-2(s-5)^{-3}\right) = 6(s-5)^{-4}$$

$$= \frac{6}{(s-5)^4}$$

$\therefore \mathcal{L}\{t^3e^{5t}\} = \frac{6}{(s-5)^4}$ (Answer)

Example-4: Find the Laplace Transform of $e^{3t}(2\cos 5t - 3\sin 5t)$

$\Rightarrow$ We have,

$$\mathcal{L}\{2\cos 5t - 3\sin 5t\} = 2\mathcal{L}\{\cos 5t\} - 3\mathcal{L}\{\sin 5t\}$$

$$2\mathcal{L}\{\cos 5t\} = 2\int_0^\infty e^{-st}\cdot\cos 5t\,.dt$$

$$= 2\left[\frac{e^{-st}}{s^2+5^2}(-s\cos 5t + 5\sin 5t)\right]_0^\infty$$

$$= 2\left(0 - \frac{1}{s^2+5^2}\cdot(-s)\right) = \frac{2s}{s^2+5^2} = \frac{2s}{s^2+25}$$

$$3\mathcal{L}\{\sin 5t\} = 3\int_0^\infty \frac{e^{-st}}{s^2+25}(-s\sin 5t - 5\cos 5t)\Big]_0^\infty$$

$$= 3\left[0 - \frac{1}{s^2+25}\times -5\right] = 3\,\frac{15}{s^2+25}$$

$\therefore$ $2\mathcal{L}\{\cos 5t\} - 3\mathcal{L}\{\sin 5t\}$

$$= \frac{2s}{s^2+25} - \frac{15}{s^2+25}$$

$$= \frac{2s-15}{s^2+25} = F(s)$$

Then by applying the first shifting theorem we have,

$$L\{e^{3t}(2\cos 5t - 3\sin 5t)\}$$

$$= f(s-3) = \frac{2(s-3) - 15}{(s-3)^2 + 25} = \frac{2s-21}{s^2-6s+34} \text{ (Answer)}$$

Definition of inverse Laplace Transform:

If $f(s)$ is the Laplace transform of a function $F(t)$, this if $L\{F(t)\} = f(s)$, then $F(t)$ is called the inverse Laplace transform of the function $f(s)$ and is symbolically written as $F(t) = L^{-1}\{f(s)\}$ where L^{-1} is called the inverse Laplace transformation operator.

Definition of Null function:

If $N(t)$ is a function of t such that,

$$\int_0^\infty N(t)\,dt = 0 \quad \text{for all} \quad t > 0.$$

Then, $N(t)$ is called a null function.

Worked Out Examples

(1) Prove that, $L^{-1}\left\{\frac{1}{s^2+a^2}\right\} = \frac{\sin at}{a}$

Proof:

$$L\left\{\frac{\sin at}{a}\right\} = \frac{1}{a} L\{\sin at\} = \frac{1}{a}\int_0^\infty e^{-st}\cdot \sin at\, dt$$

$$= \frac{1}{a}\left[\frac{e^{-st}}{s^2+a^2}(-s\sin at - a\cos at)\right]_0^\infty$$

$$= \frac{1}{a}\left(0 - \frac{1}{s^2+a^2}\times -a\right) = \frac{1}{a}\times\frac{a}{s^2+a^2} = \frac{1}{s^2+a^2}$$

$$\therefore \frac{\sin at}{a} = L^{-1}\left\{\frac{1}{s^2+a^2}\right\} \text{ (proved)}$$

(11) Prove that, $L^{-1}\left\{\frac{1}{s^{n+1}}\right\} = \frac{t^n}{\lfloor n+1}, n > 1$

Proof:

$$L\left\{\frac{t^n}{\lfloor n+1}\right\} = \frac{1}{\lfloor n+1} L\{t^n\}$$

$$= \frac{1}{\lfloor n+1}\int_0^\infty e^{-st} t^n\, dt \quad \text{---(1)}$$

Putting $st = y$, so that $s\,dt = dy \therefore dt = \frac{dy}{s}$

Also, $t^n = \left(\frac{y}{s}\right)^n$ Limits $\begin{matrix} t=\infty \\ y=\infty \end{matrix}\Big\}\ \begin{matrix} t=0 \\ y=0 \end{matrix}\Big\}$

Thus from (i) we have,

$$= \frac{1}{\lfloor n+1} \int_0^\infty e^{-y} \cdot \frac{y^n}{s^n} \cdot \frac{dy}{s}$$

$$= \frac{1}{\lfloor n+1} \cdot \frac{1}{s^{n+1}} \int_0^\infty e^{-y} \cdot y^n dy \quad \text{(using gamma function)}$$

$$= \frac{1}{\lfloor n+1} \cdot \frac{\lfloor n+1}{s^{n+1}} = \frac{1}{s^{n+1}}$$

$\therefore L^{-1}\left\{\frac{1}{s^{n+1}}\right\} = \frac{t^n}{\lfloor n+1}$ (proved)

Prove that, $L^{-1}\left\{\frac{1}{(s-a)^2+b^2}\right\} = \frac{1}{b} e^{at} \sin bt$

$\Rightarrow L\left\{\frac{1}{b} e^{at} \sin bt\right\}$

$= \frac{1}{b} L\{e^{at} \sin bt\}$

$= \frac{1}{b} \int_0^\infty e^{-st} e^{at} \sin bt \, dt$

$= \frac{1}{b} \int_0^\infty (e^{-t(s-a)} \sin bt) dt$

$= \frac{1}{b} \left[\frac{e^{-t(s-a)}}{(s-a)^2+b^2} \left(-(s-a)\sin bt - b\cos bt \right) \right]_0^\infty$

$= \frac{1}{b} \left(0 - \frac{1}{(s-a)^2+b^2} \times (-b) \right)$

$= \frac{1}{b} \times \frac{b}{(s-a)^2+b^2} = \frac{1}{(s-a)^2+b^2}$

$\therefore \frac{1}{b} e^{at} \sin bt = L^{-1}\left\{\frac{1}{(s-a)^2+b^2}\right\}$ (proved)

Prove that, $L^{-1}\left\{\frac{s-a}{(s-a)^2+b^2}\right\} = e^{at}\cos bt$

Proof:-

$$L\{e^{at}\cos bt\}$$

$$= \int_0^\infty e^{-st} e^{at} \cos bt \, dt$$

$$= \int_0^\infty e^{-t(s-a)} \cos bt \, dt$$

$$= \left[\frac{e^{-t(s-a)}}{(s-a)^2+b^2}\left(-(s-a)\cos bt + b\sin bt\right)\right]_0^\infty$$

$$= 0 - \frac{1}{(s-a)^2+b^2} \times (-(s-a))$$

$$= \frac{s-a}{(s-a)^2+b^2}$$

$\therefore e^{at}\cos bt = L^{-1}\left\{\frac{s-a}{(s-a)^2+b^2}\right\}$

(Proved)

Division by t

If $L\{F(t)\} = f(s)$ then,

$$L\left\{\frac{F(t)}{t}\right\} = \int_s^\infty f(u)\,du$$

Provided $\lim_{t\to 0} \frac{F(t)}{t}$ exists.

* Show that,

$$L\left\{\frac{e^{-at} - e^{-bt}}{t}\right\} = \ln\left(\frac{s+b}{s+a}\right)$$

Solution

Let, $F(t) = e^{-at} - e^{-bt}$

$$L\{F(t)\} = L\{e^{-at} - e^{-bt}\}$$

$$= \frac{1}{s+a} - \frac{1}{s+b} = f(s)$$

We know that,

$$L\left\{\frac{F(t)}{t}\right\} = \int_s^\infty f(u)\,du,$$

$$\therefore L\left\{\frac{e^{-at} - e^{-bt}}{t}\right\} = \int_s^\infty \left(\frac{1}{u+a} - \frac{1}{u+b}\right) du$$

$$= \left[\ln|u+a| - \ln|u+b|\right]_s^\infty$$

$$= \ln|\infty+a| - \ln|\infty+b| - \ln|s+a| + \ln|s+b|$$

$$= \ln\infty - \ln\infty + \ln\left|\frac{s+b}{s+a}\right|$$

$$= \ln\left|\frac{s+b}{s+a}\right|$$

(Showed)

Show that,

$$L\left\{\frac{\cos at - \cos bt}{t}\right\} = \frac{1}{2}\ln\left|\frac{s^2+b^2}{s^2+a^2}\right|$$

⇒ Here,

$$L\left\{\frac{\cos at - \cos bt}{t}\right\}$$

Division by t Rule

If $L\{F(t)\} = f(s)$ then

$$L\left\{\frac{F(t)}{t}\right\} = \int_s^\infty F(u)\,du$$

Here, $F(t) = \cos at - \cos bt$

We know,

$$L\{F(t)\} = L\{\cos at\} = \frac{s}{s^2+a^2}, \text{ here } F(t) = \cos at$$

$$L\{F(t)\} = L\{\cos bt\} = \frac{s}{s^2+b^2}, \quad F(t) = \cos bt$$

$$f(s) = L\{F(t)\} = \frac{s}{s^2+a^2} - \frac{s}{s^2+b^2}$$

Now;

$$L\left\{\frac{F(t)}{t}\right\} = L\left\{\frac{\cos at - \cos bt}{t}\right\}$$

$$= \int_s^\infty \left[\frac{u}{u^2+a^2} - \frac{u}{u^2+b^2}\right] du$$

$$= \left[\frac{1}{2}\ln(u^2+a^2) - \frac{1}{2}\ln(u^2+b^2)\right]_s^\infty$$

At $u \to \infty$, the logarithmic tends to 0.

So,

$$= \frac{1}{2}\ln(s^2+a^2) + \frac{1}{2}\ln(s^2+b^2)$$

$$= \frac{1}{2}\left[\ln\left(\frac{s^2+b^2}{s^2+a^2}\right)\right]$$

(Showed)

Showed that,

$$\int_0^\infty \frac{e^{-3t} - e^{-6t}}{t}\,dt = \ln 2$$

$\Rightarrow$ We know that,

If $L\{F(t)\} = f(s)$

then, $L\left\{\frac{F(t)}{t}\right\} = \int_s^\infty F(u)\,du$

Here,

$$F(t) = e^{-3t} - e^{-6t}$$

$$L\{F(t)\} = L\{e^{-3t}\} = \frac{1}{s+3}, \quad F(t) = e^{-3t}$$

$$L\{F(t)\} = L\{e^{-6t}\} = \frac{1}{s+6}, \quad F(t) = e^{-6t}$$

$$\therefore L\{F(t)\} = \frac{1}{s+3} - \frac{1}{s+6}$$

Now, $S=0$ then

$$\int_0^\infty \frac{e^{-3t}-e^{-6t}}{t}\,dt = \int_0^\infty \left(\frac{1}{S+3}-\frac{1}{S+6}\right) dS$$

$$= \left[\ln(S+3) - \ln(S+6)\right]_0^\infty$$

$S \to \infty$ then

$$\ln \frac{S+3}{S+6} \to \ln 1 = 0$$

$S \to 0$ then

$$\ln \frac{3}{6} = \ln \frac{1}{2}$$

$$\therefore 0 - \ln \frac{1}{2} = \ln 2$$

(Showed)